

Ultra-Broadband SCVM Phase Shifter

Ideal Mathematical Model Derivations

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§1 Introduction

Below are my expanded step-by-step derivations for the mathematical models of the novel Self-Compensated Vector Modulation (SCVM) Phase Shifter found in [JSSC 2025](#). I summarize the main mathematical ideas in [section 6](#).

§2 Vector-Modulation Phase Shifter

The voltage gain in the in-phase (I) and quadrature (Q) paths of a Lange coupler with center frequency f_C and coupling coefficient C are

$$I = \frac{\sqrt{1 - C^2}}{\sqrt{1 - C^2} \cos\left(\frac{\pi f}{2f_C}\right) + j \sin\left(\frac{\pi f}{2f_C}\right)}$$
$$Q = \frac{jC \sin\left(\frac{\pi f}{2f_C}\right)}{\sqrt{1 - C^2} \cos\left(\frac{\pi f}{2f_C}\right) + j \sin\left(\frac{\pi f}{2f_C}\right)}.$$

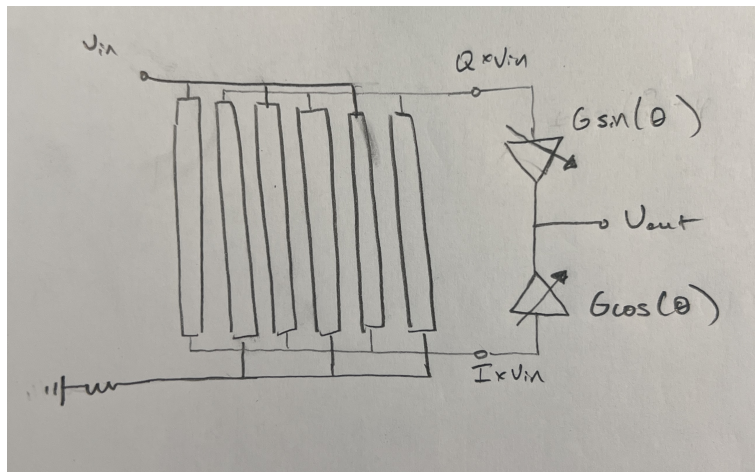


Figure 1: One-stage Vector Modulated Phase Shifter

By setting the gains of the two VGAs as $G \cos(\theta)$ and $G \sin(\theta)$, we get

$$\begin{aligned} V_{\text{out}} &= IV_{\text{in}} \cdot G \cos(\theta) + QV_{\text{in}} \cdot G \sin(\theta) \\ &= \left[\frac{(\cos(\theta))\sqrt{1-C^2} + jC \sin\left(\frac{\pi f}{2f_C}\right) \sin(\theta)}{\sqrt{1-C^2} \cos\left(\frac{\pi f}{2f_C}\right) + j \sin\left(\frac{\pi f}{2f_C}\right)} \right] GV_{\text{in}}. \end{aligned}$$

Dividing the numerator and denominator by $\sqrt{1-C^2}$, we get

$$V_{\text{out}} = \left[\frac{\cos(\theta) + j \frac{C}{\sqrt{1-C^2}} \sin\left(\frac{\pi f}{2f_C}\right) \sin(\theta)}{\cos\left(\frac{\pi f}{2f_C}\right) + \frac{j \sin\left(\frac{\pi f}{2f_C}\right)}{\sqrt{1-C^2}}} \right] GV_{\text{in}}.$$

Letting $G = \left(\cos\left(\frac{\pi f}{2f_C}\right) + \frac{j \sin\left(\frac{\pi f}{2f_C}\right)}{\sqrt{1-C^2}} \right) G_0$, we can substitute in for G to cancel the denominator and get

$$V_{\text{out}} = \left(\cos(\theta) + j \frac{C}{\sqrt{1-C^2}} \sin\left(\frac{\pi f}{2f_C}\right) \sin(\theta) \right) G_0 V_{\text{in}}.$$

From this equation, we can see the problems with the VM-PS. For varying frequencies and θ , the phase shifting and magnitude are not consistent. The authors propose the use of N cascaded stages of VM-PS to solve this problem.

§3 Ideal Self-Compensated-Vector-Modulation Phase Shifter

The SCVM idea is to divide the total phase shift into N equal parts and give each an initial phase offset of φ_k at equal intervals from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. This will lead to the "self-compensation" idea.

§3.1 Getting the Cascaded Gain Response

Let $\varphi_k = \frac{2k-1-N}{2N}\pi$, with $k \in [1, N]$. Respectively, each phase shifter will do a phase shift of

$$\frac{\theta}{N} + \varphi_1, \frac{\theta}{N} + \varphi_2, \dots, \frac{\theta}{N} + \varphi_N.$$

Since each phase shifter is a VM-PS, the k th phase shifter has a gain of

$$\begin{aligned} G_k &= I \cdot G \cos\left(\frac{\theta}{N} + \varphi_k\right) + Q \cdot G \sin\left(\frac{\theta}{N} + \varphi_k\right) \\ &= \left(I \cos\left(\frac{\theta}{N} + \varphi_k\right) + Q \sin\left(\frac{\theta}{N} + \varphi_k\right) \right) G. \end{aligned}$$

Thus, the cascaded gain response of the whole phase shifter G_{PS} can be written as

$$G_{\text{PS}} = G^N \prod_{k=1}^N \left[I \cos\left(\frac{\theta}{N} + \varphi_k\right) + Q \sin\left(\frac{\theta}{N} + \varphi_k\right) \right].$$

For simplicity, let $\alpha_k = \frac{\theta}{N} + \varphi_k$. Since $\cos(\theta) = \frac{\text{cis}(\theta) + \text{cis}(-\theta)}{2}$ and $\sin(\theta) = \frac{j \text{cis}(-\theta) - j \text{cis}(\theta)}{2}$, We can rewrite the sum inside the product as

$$\begin{aligned} I \cos(\alpha_k) + Q \sin(\alpha_k) &= \frac{I}{2} (\text{cis}(\alpha_k) + \text{cis}(-\alpha_k)) + \frac{Q}{2} (j \text{cis}(-\alpha_k) - j \text{cis}(\alpha_k)) \\ &= \frac{I - jQ}{2} \text{cis}(\alpha_k) + \frac{I + jQ}{2} \text{cis}(-\alpha_k) \\ &= \frac{I - jQ}{2} e^{j\alpha_k} + \frac{I + jQ}{2} e^{-j\alpha_k}. \end{aligned}$$

Thus, we can write

$$\begin{aligned} G_{\text{PS}} &= G^N \prod_{k=1}^N \left(\frac{I - jQ}{2} e^{j\alpha_k} + \frac{I + jQ}{2} e^{-j\alpha_k} \right) \\ &= G^N \prod_{k=1}^N \left(\frac{I - jQ}{2} e^{j\alpha_k} \right) \left(1 + \frac{I + jQ}{I - jQ} e^{-2j\alpha_k} \right) \\ &= G^N \prod_{k=1}^N \left(\frac{I - jQ}{2} e^{j\alpha_k} \right) \prod_{k=1}^N \left(1 + \frac{I + jQ}{I - jQ} e^{-2j\alpha_k} \right) \end{aligned}$$

Plugging back in $\alpha_k = \frac{\theta}{N} + \varphi_k$,

$$\begin{aligned} G_{\text{PS}} &= G^N \prod_{k=1}^N \left(\frac{I - jQ}{2} e^{j(\frac{\theta}{N} + \varphi_k)} \right) \prod_{k=1}^N \left(1 + \frac{I + jQ}{I - jQ} e^{-2j\alpha_k} \right) \\ &= G^N \prod_{k=1}^N \left(\frac{I - jQ}{2} e^{j\frac{\theta}{N}} \right) \prod_{k=1}^N (e^{j\varphi_k}) \prod_{k=1}^N \left(1 + \frac{I + jQ}{I - jQ} e^{-2j\alpha_k} \right). \end{aligned}$$

Since we know that

$$\varphi_k = \frac{2k - 1 - N}{2N} \pi = -\frac{N + 1 - 2k}{2N} \pi = -\frac{2(N - k + 1) - 1 - N}{2N} \pi = -\varphi_{N-k+1},$$

by symmetry, $\sum_{k=1}^N \varphi_k = 0$.

Thus,

$$\prod_{k=1}^N e^{j\varphi_k} = e^{j \sum_{k=1}^N \varphi_k} = e^0 = 1.$$

So we have

$$G_{\text{PS}} = G^N \prod_{k=1}^N \left(\frac{I - jQ}{2} e^{j\frac{\theta}{N}} \right) \prod_{k=1}^N \left(1 + \frac{I + jQ}{I - jQ} e^{-2j\alpha_k} \right).$$

§3.2 Clearing up the Second Product

Now, we would like to clear up the second product term. Define

$$x_k = \frac{I + jQ}{I - jQ} e^{-2j\alpha_k}.$$

Then we can rewrite x_k as

$$\begin{aligned} x_k &= \left(\frac{I + jQ}{I - jQ} \right) e^{-2j\alpha_k} \\ &= \left(\frac{I + jQ}{I - jQ} \right) e^{-2j(\frac{\theta}{N} + \varphi_k)} \\ &= \left(\frac{I + jQ}{I - jQ} \right) e^{-2j(\frac{\theta}{N})} e^{j(-2\varphi_k)} \\ &= \left(\frac{I + jQ}{I - jQ} \right) e^{-2j(\frac{\theta}{N})} e^{j\frac{N+1-2k}{N}\pi} \\ &= \left(\frac{I + jQ}{I - jQ} \right) e^{-2j(\frac{\theta}{N})} e^{j\pi} e^{j\frac{\pi}{N}} e^{-j\frac{2k\pi}{N}} \\ &= \left(-\frac{I + jQ}{I - jQ} \right) e^{-2j(\frac{\theta}{N})} e^{j\frac{\pi}{N}} e^{-j\frac{2k\pi}{N}} \end{aligned}$$

Taking both sides to the power of N ,

$$\begin{aligned} x_k^N &= \left(-\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta} e^{j\pi} e^{-j(2k\pi)} \\ &= \left(-\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta} (-1)(1) \\ &= - \left(-\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta} \end{aligned}$$

So, we have

$$x_k^N + \left(-\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta} = 0.$$

Now, we can see that x_k form exactly the N solutions to

$$x^N + \left(-\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta} = 0.$$

By Vieta's formulas, we see that all its elementary symmetric polynomials are equal to zero except for the product of all the roots. Furthermore, Vieta's also tells us

$$\prod_{k=1}^N x_k = (-1)^N \left(-\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta} = \left(\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta}.$$

Thus,

$$\begin{aligned}
\prod_{k=1}^N \left(1 + \frac{I + jQ}{I - jQ} e^{-2j\alpha_k} \right) &= \prod_{k=1}^N (1 + x_k) \\
&= 1 + \sum_{i_1} x_{i_1} + \sum_{i_1 < i_2} x_{i_1} x_{i_2} + \cdots + \prod_{k=1}^N x_k \\
&= 1 + \prod_{k=1}^N x_k \\
&= 1 + \left(\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta}
\end{aligned}$$

This allows us to simplify G_{PS} to

$$\begin{aligned}
G_{\text{PS}} &= G^N \prod_{k=1}^N \left(\frac{I - jQ}{2} e^{j\frac{\theta}{N}} \right) \prod_{k=1}^N \left(1 + \frac{I + jQ}{I - jQ} e^{-2j\alpha_k} \right) \\
&= G^N \prod_{k=1}^N \left(\frac{I - jQ}{2} e^{j\frac{\theta}{N}} \right) \left(1 + \left(\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta} \right) \\
&= G^N \left(\frac{I - jQ}{2} \right)^N (e^{j\theta}) \left(1 + \left(\frac{I + jQ}{I - jQ} \right)^N e^{-2j\theta} \right) \\
&= \left(\frac{I - jQ}{2} G \right)^N (e^{j\theta}) \left(1 + \left(\frac{I + jQ}{I - jQ} \right)^N e^{-j2\theta} \right).
\end{aligned}$$

§3.3 Error Term and Self-Compensation

Now, define

$$\Delta_{\text{err}} = \left(\frac{I + jQ}{I - jQ} \right)^N e^{-j2\theta}.$$

This gives

$$G_{\text{PS}} = \left(\frac{I - jQ}{2} G \right)^N e^{j\theta} (1 + \Delta_{\text{err}}).$$

In an ideal case, $Q = jI$ and $I + jQ = 0$. Thus, the Δ_{err} term will disappear in this case.

However, the amplitudes of I and Q will not necessarily match, and they will not necessarily be perfectly orthogonal, leading this $I + jQ$ term to arise.

For a small mismatch, let $Q = jI + \epsilon$ for a small ϵ . Then

$$\begin{aligned}\Delta_{\text{err}} &= \left(\frac{I + jQ}{I - jQ} \right)^N e^{-j2\theta} \\ &= \left(\frac{I + (-I + \epsilon j)}{I + I - \epsilon j} \right)^N e^{-j2\theta} \\ &= \underbrace{\left(\frac{\epsilon j}{2I - \epsilon j} \right)^N}_{<1} e^{-j2\theta}\end{aligned}$$

Thus, we see that as N increases, the Δ_{err} will go to 0.

Furthermore, as N increases, G_{PS} will approach $\left(\frac{I - jQ}{2} G \right)^N e^{j\theta}$. Thus, we see the total phase shift of θ .

§3.4 Phase and Magnitude Errors

We can define the phase and magnitude errors as

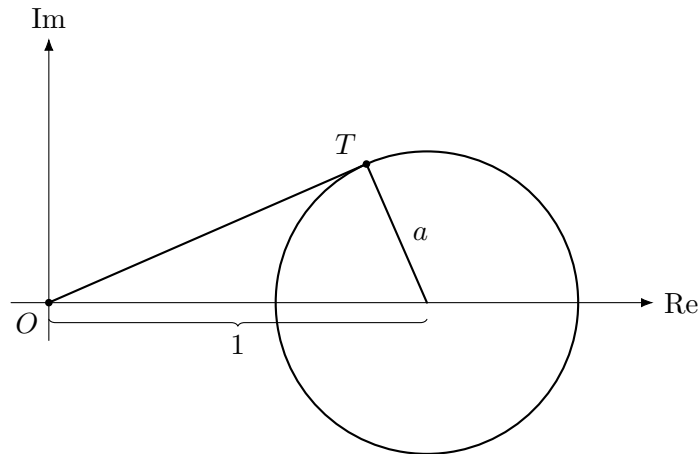
$$\begin{aligned}\Delta_{\theta} &= \text{phase}(1 + \Delta_{\text{err}}) = \text{phase} \left[1 + \left(\frac{I + jQ}{I - jQ} \right)^N e^{-j2\theta} \right] \\ \Delta_G &= \text{dB}(1 + \Delta_{\text{err}}) = \text{dB} \left[1 + \left(\frac{I + jQ}{I - jQ} \right)^N e^{-j2\theta} \right].\end{aligned}$$

When $\Delta_{\text{err}} = 0$, both $\Delta_{\theta}, \Delta_G = 0$. For the sake of simplicity, let

$$\begin{aligned}\psi &= \text{phase} \left(\left(\frac{I + jQ}{I - jQ} \right)^N e^{-j2\theta} \right) \\ a &= \left| \frac{I + jQ}{I - jQ} \right|^N\end{aligned}$$

Then, $1 + \Delta_{\text{err}} = 1 + ae^{j\psi}$, $\Delta_{\theta} = \text{phase}(1 + ae^{j\psi})$, and $\Delta_G = \text{dB}(1 + ae^{j\psi})$.

This allows us to analyze $1 + \Delta_{\text{err}}$ on the complex plane. Here, all $1 + \Delta_{\text{err}}$ lie on a circle centered at $(1, 0)$ with radius a .



We would like to find the maximum phase and magnitude errors. The maximum phase error is at point T (and symmetrically at a T' that is T reflected over the real axis).

Thus,

$$\begin{aligned} |\Delta_\theta|_{\max} &= \arcsin(a) \\ &= \arcsin\left(\left|\frac{I+jQ}{I-jQ}\right|^N\right). \end{aligned}$$

In a similar way, we can find $|\Delta_G|_{\max}$. From our figure, we can see that

$$1 - a \leq |1 + \Delta_{\text{err}}| \leq 1 + a.$$

Since $1 - a < 1$, we can write

$$\begin{aligned} |\Delta_G|_{\max} &= \max\{|\text{dB}(1 - a)|, |\text{dB}(1 + a)|\} \\ &= \max\{-\text{dB}(1 - a), \text{dB}(1 + a)\}. \end{aligned}$$

Comparing the two by subtraction,

$$-\text{dB}(1 - a) - \text{dB}(1 + a) = -(\text{dB}(1 - a^2)).$$

Because $0 < 1 - a^2 < 1$, we know that $-(\text{dB}(1 - a^2)) = -20 \log_{10}(1 - a^2) > 0$.

Thus, $-\text{dB}(1 - a)$ is larger and

$$\begin{aligned} |\Delta_G|_{\max} &= \max\{-\text{dB}(1 - a), \text{dB}(1 + a)\} \\ &= -\text{dB}(1 - a) \\ &= -\text{dB}\left(1 - \left|\frac{I+jQ}{I-jQ}\right|^N\right). \end{aligned}$$

Here, we can again see how increasing N will significantly reduce the phase and magnitude errors.

§3.5 Ensuring Ultra-Broadband by bounding $|I - jQ|$

We know that

$$G_{\text{PS}} = \left(\frac{I - jQ}{2}G\right)^N e^{j\theta} (1 + \Delta_{\text{err}}).$$

We would like $\left(\frac{I - jQ}{2}G\right)^N$ to remain stable for a large bandwidth, so we will bound $|I - jQ|$. In an ideal Lange coupler,

$$|I - jQ| = \left| \frac{\sqrt{1 - C^2} + C \sin\left(\frac{\pi f}{2f_C}\right)}{\sqrt{1 - C^2} \cos\left(\frac{\pi f}{2f_C}\right) + j \sin\left(\frac{\pi f}{2f_C}\right)} \right|.$$

For the sake of simplicity, let $x = \frac{\pi f}{2f_C}$. Then,

$$\begin{aligned}
 |I - jQ| &= \left| \frac{\sqrt{1 - C^2} + C \sin(x)}{\sqrt{1 - C^2} \cos(x) + j \sin(x)} \right| \\
 &= \frac{|\sqrt{1 - C^2} + C \sin(x)|}{|\sqrt{1 - C^2} \cos(x) + j \sin(x)|} \\
 &= \frac{|\sqrt{1 - C^2} + C \sin(x)|}{\sqrt{(1 - C^2) \cos^2(x) + \sin^2(x)}} \\
 &= \sqrt{\frac{(\sqrt{1 - C^2} + C \sin(x))^2}{(1 - C^2) \cos^2(x) + \sin^2(x)}}
 \end{aligned}$$

Expanding and rewriting $(1 - C^2) \cos^2(x) + \sin^2(x)$ as $(1 - C^2) + C^2 \sin^2(x)$, we get

$$\begin{aligned}
 |I - jQ| &= \sqrt{\frac{(1 - C^2) + C^2 \sin^2(x) + 2C\sqrt{1 - C^2} \sin(x)}{(1 - C^2) + C^2 \sin^2(x)}} \\
 &= \sqrt{1 + \frac{2C\sqrt{1 - C^2} \sin(x)}{(1 - C^2) + C^2 \sin^2(x)}} \\
 &= \sqrt{1 + \frac{2C\sqrt{1 - C^2}}{\frac{1 - C^2}{\sin(x)} + C^2 \sin(x)}}.
 \end{aligned}$$

Looking at $\frac{1 - C^2}{\sin(x)} + C^2 \sin(x)$, we see that it is minimized and equal to 1 when $x = \pi/2$. Furthermore, since in the case of an ideal Lange Coupler, $|I|, |Q| \leq 1$, we know that $|I - jQ| \leq |I| + |Q| \leq 2$. Thus,

$$1 < |I - jQ| \leq \min\{\sqrt{1 + 2C\sqrt{1 - C^2}}, 2\}.$$

So, if we loosely bound $|I - jQ|/2$ as $0.5 < |I - jQ|/2 \leq 1$, we find that in the worst case,

$$\left| \text{dB} \left(\frac{|I - jQ|}{2} \right) \right| = \left| 20 \log_{10} \left(\frac{1}{2} \right) \right| \approx 6.$$

Thus, the gain fluctuation is less than 6 dB in each stage.

§4 Impact of Parasitics and Distortion

Because of non-idealities, the gain of the VGA can be expressed as

$$G_{\text{VGA}} = A(V_{\text{ctrl}})H(f)[1 + \delta(A, f)]$$

where $A(V_{\text{ctrl}})$ represents the idea target gain control profiles, $H(f)$ represents the bias-independent frequency response, and $\delta(A, f)$ is the distortion factor accounting for gain and phase variation from parasitic and non-linearity.

Because we can design VGA to be symmetric, $\delta(A, f)$ is an even function with respect to A and it reduces to 0 when $A(V_{\text{ctrl}}) = 0$. Thus,

$$\delta(0, f) = 0, \delta(A, f) = \delta(-A, f).$$

§4.1 Gain of Phase Shifter

The gains in the in-phase and quadrature paths will be

$$\begin{aligned} G_{I,k} &= A_0 \cos(\alpha_k) H(f) (1 + \delta(A_0 \cos(\alpha_k), f)), \\ G_{Q,k} &= A_0 \sin(\alpha_k) H(f) (1 + \delta(A_0 \sin(\alpha_k), f)). \end{aligned}$$

Thus, the gain in the phase shifter G_{PS} is

$$G_{\text{PS}} = A_0^N H^N(f) \prod_{k=1}^N \left\{ \begin{array}{l} I \cos(\alpha_k) (1 + \delta(A_0 \cos(\alpha_k), f)) \\ + Q \sin(\alpha_k) (1 + \delta(A_0 \sin(\alpha_k), f)) \end{array} \right\}.$$

Doing a Taylor expansion on $\delta(A, f)$ around $A = 0$, we find

$$\delta(A, f) = \delta_0(f) + \delta_1(f)A + \delta_2(f)A^2 + \delta_3(f)A^3 + \dots,$$

where $\delta_k(f) = \frac{1}{k!} \frac{\partial^k \delta}{\partial A^k}(A, f) \Big|_{A=0}$.

We know that $\delta_0(f) = \delta(0, f) = 0$. Considering $\delta(A, f)$ and $\delta(-A, f)$ now, we see that

$$\begin{aligned} \delta(A, f) &= \delta_0(f) + \delta_1(f)A + \delta_2(f)A^2 + \delta_3(f)A^3 + \dots \\ \delta(-A, f) &= \delta_0(f) - \delta_1(f)A + \delta_2(f)A^2 - \delta_3(f)A^3 + \dots \end{aligned}$$

Since we know $\delta(A, f) = \delta(-A, f)$, we see that for odd k , $\delta_k(f)$ must be 0. Thus, it's reasonable to approximate

$$\delta(A, f) \approx \delta_2 A^2.$$

Define $G_{\text{norm}} = \frac{G_{\text{PS}}}{A_0^N H^N(f)}$, and take the approximation $\delta(A, f) \approx \delta_2 A^2$. Then,

$$G_{\text{norm}} = \prod_{k=1}^N \left\{ \begin{array}{l} I \cos(\alpha_k) + I \delta_2 A_0^2 \cos^3(\alpha_k) \\ + Q \sin(\alpha_k) + Q \delta_2 A_0^2 \sin^3(\alpha_k) \end{array} \right\}.$$

Now we can utilize the triple angle identities

$$\cos(3\theta) = 4 \cos^3(\theta) - 3 \cos(\theta)$$

$$\sin(3\theta) = 3 \sin(\theta) - 4 \sin^3(\theta).$$

Thus, we find that

$$\begin{aligned} G_{\text{norm}} &= \prod_{k=1}^N \left\{ \begin{aligned} &I \cos \alpha_k + I \delta_2 A_0^2 \left(\frac{\cos(3\alpha_k) + 3 \cos \alpha_k}{4} \right) \\ &+ Q \sin \alpha_k + Q \delta_2 A_0^2 \left(\frac{3 \sin \alpha_k - \sin(3\alpha_k)}{4} \right) \end{aligned} \right\} \\ &= \prod_{k=1}^N \left\{ \begin{aligned} &I \cos(\alpha_k) \left(1 + \frac{3}{4} \delta_2 A_0^2 \right) + \frac{1}{4} I \delta_2 A_0^2 \cos(3\alpha_k) \\ &+ Q \sin(\alpha_k) \left(1 + \frac{3}{4} \delta_2 A_0^2 \right) - \frac{1}{4} Q \delta_2 A_0^2 \sin(3\alpha_k) \end{aligned} \right\}. \end{aligned}$$

Since $I \cos(\theta) + Q \sin(\theta) = \frac{I-jQ}{2} \text{cis}(\theta) + \frac{I+jQ}{2} \text{cis}(\theta)$ and similarly for $I \cos(\theta) - Q \sin(\theta)$,

$$\begin{aligned} G_{\text{norm}} &= \prod_{k=1}^N \left\{ \begin{aligned} &\frac{I-jQ}{2} \left(1 + \frac{3}{4} \delta_2 A_0^2 \right) e^{j\alpha_k} + \frac{I+jQ}{2} \left(1 + \frac{3}{4} \delta_2 A_0^2 \right) e^{-j\alpha_k} \\ &+ \frac{I+jQ}{2} \left(\frac{1}{4} \delta_2 A_0^2 \right) e^{j3\alpha_k} + \frac{I-jQ}{2} \left(\frac{1}{4} \delta_2 A_0^2 \right) e^{-j3\alpha_k} \end{aligned} \right\} \\ &= \left(\frac{I-jQ}{2} \cdot \frac{\delta_2 A_0^2}{4} \right)^N \prod_{k=1}^N e^{j3\alpha_k} \prod_{k=1}^N \left\{ \begin{aligned} &\frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} e^{-j2\alpha_k} + \frac{I+jQ}{I-jQ} \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} e^{-j4\alpha_k} \\ &+ \frac{I+jQ}{I-jQ} + e^{-j6\alpha_k} \end{aligned} \right\}. \end{aligned}$$

Also, since

$$\prod_{k=1}^N e^{j3\alpha_k} = \prod_{k=1}^N e^{j(3\theta/N)} \prod_{k=1}^N e^{j\varphi_k} = e^{j3\theta},$$

we find that

$$G_{\text{norm}} = e^{j3\theta} \left(\frac{I-jQ}{2} \cdot \frac{\delta_2 A_0^2}{4} \right)^N \prod_{k=1}^N \left\{ \begin{aligned} &\frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} e^{-j2\alpha_k} + \frac{I+jQ}{I-jQ} \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} e^{-j4\alpha_k} \\ &+ \frac{I+jQ}{I-jQ} + e^{-j6\alpha_k} \end{aligned} \right\}.$$

Define the third order Laurent polynomial $L(x)$ where

$$L(x) = x^3 + \frac{I+jQ}{I-jQ} \cdot \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} x^2 + \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} x + \frac{I+jQ}{I-jQ}.$$

Then we can write

$$G_{\text{norm}} = e^{j3\theta} \left(\frac{I-jQ}{2} \cdot \frac{\delta_2 A_0^2}{4} \right)^N \prod_{k=1}^N L(e^{-2j\alpha_k}).$$

Let the three roots of $L(x)$ be x_1, x_2, x_3 where $|x_1| \leq |x_2| \leq |x_3|$. Then,

$$G_{\text{norm}} = e^{j3\theta} \left(\frac{I-jQ}{2} \cdot \frac{\delta_2 A_0^2}{4} \right)^N \prod_{m=1}^3 \prod_{k=1}^N (e^{\alpha_k} - x_m).$$

To simplify each $\prod_{k=1}^N (e^{\alpha_k} - x_m)$, we proceed similarly as in [subsection 3.2](#).

Let $y_k = e^{-2\alpha_k}$. Then,

$$\begin{aligned} y_k &= e^{-2\alpha_k} \\ &= e^{-j\frac{2\theta}{N} - j2\varphi_k} \\ &= e^{-j\frac{2\theta}{N}} e^{-j2\varphi_k} \\ &= -e^{-j\frac{2\theta}{N}} e^{j\frac{\pi}{N}} e^{-j\frac{2k\pi}{N}}. \end{aligned}$$

Then, considering y_k^N , we find that

$$\begin{aligned} y_k^N &= (-1)^N e^{-j2\theta} \left(e^{j\frac{\pi}{N}} e^{-j\frac{2k\pi}{N}} \right) \\ &= -(-e^{-j2\theta})^N. \end{aligned}$$

Thus, y_k form exactly the N solutions to

$$y^N + (-e^{-j2\theta})^N = 0.$$

Like in [subsection 3.2](#), applying Vieta's tells us that of the elementary symmetric polynomials, only the product of all the roots is nonzero. We know that this product is equal to

$$\prod_{k=1}^N y_k = e^{-2j\theta}.$$

Thus, we find that

$$\begin{aligned} \prod_{k=1}^N (y_k - x_m) &= \left\{ (-x_m)^N + (-x_m)^{N-1} \left(\sum_{i_1} y_{i_1} \right) + (-x_m)^{N-2} \left(\sum_{i_1 < i_2} y_{i_1} y_{i_2} \right) \right. \\ &\quad \left. + (-x_m)^{N-3} \left(\sum_{i_1 < i_2 < i_3} y_{i_1} y_{i_2} y_{i_3} \right) + \cdots + \prod_{k=1}^N y_k \right\} \\ &= (-x_m)^N + \prod_{k=1}^N y_k \\ &= (-x_m)^N + e^{-2j\theta}. \end{aligned}$$

Thus,

$$G_{\text{norm}} = e^{j3\theta} \left(\frac{I - jQ}{2} \frac{\delta_2 A_0^2}{4} \right)^N [e^{-j2\theta} + (-x_1)^N] [e^{-j2\theta} + (-x_2)^N] [e^{-j2\theta} + (-x_3)^N].$$

This shows that x_1, x_2, x_3 are important for determining the response of the phase shifter. The authors assume the following:

1. The signal in Q -path leads the signal in the I path, $0 < \theta(Q) - \theta(I) < \pi$, implying that $\left| \frac{I+jQ}{I-jQ} \right| < 1$.
2. Distortion is not severe, implying that $|\delta_2 A_0^2|$ is small, so $\left| \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} \right| \gg 1$.

§4.2 Locating the roots

We defined $L(x)$ to be

$$L(x) = x^3 + \frac{I+jQ}{I-jQ} \cdot \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} x^2 + \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} x + \frac{I+jQ}{I-jQ}.$$

Define $f(x), g(x)$ where $f(x) + g(x) = L(x)$ as follows:

$$\begin{aligned} f(x) &= \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} x \\ g(x) &= x^3 + \frac{I+jQ}{I-jQ} \cdot \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} x^2 + \frac{I+jQ}{I-jQ}. \end{aligned}$$

Recall that we defined this Laurent polynomial with $x = e^{-2j\alpha_k}$, so we would like to analyze x for $|x| = 1$. Thus, we have

$$\begin{aligned} |f(x)| &= \left| \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} \right| \\ |g(x)| &\leq 1 + \left| \frac{I+jQ}{I-jQ} \right| \left| \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} \right| + \left| \frac{I+jQ}{I-jQ} \right|. \end{aligned}$$

Continuing with the assumptions that Q leads I and distortion is not severe, we find that

$$\begin{aligned} |f(x)| - |g(x)| &\geq \left| \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} \right| \left(1 - \left| \frac{I+jQ}{I-jQ} \right| \right) - \left| \frac{I+jQ}{I-jQ} \right| - 1 \\ &> 0. \end{aligned}$$

Thus, $|f(x)| > |g(x)|$ for all $|x| = 1$ on the boundary of the unit circle.

By Rouché's theorem, since $f(x)$ has only one zero root ($x = 0$) inside the unit circle $|x| < 1$, we know that $L(x) = f(x) + g(x)$ should only have one zero root in $|x| < 1$.

To locate the other two roots, we proceed in a similar fashion as above. Define $y = 1/x$ and write

$$\begin{aligned} Q(y) &= y^3 L\left(\frac{1}{y}\right) \\ &= 1 + \frac{I+jQ}{I-jQ} \cdot \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} y + \frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} y^3 + \frac{I+jQ}{I-jQ} y^3. \end{aligned}$$

Then, define $F(y)$, $G(y)$ where $F(y) + G(y) = Q(y)$ as follows:

$$F(x) = \frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} y^2$$

$$G(x) = 1 + \frac{I + jQ}{I - jQ} \cdot \frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} y + \frac{I + jQ}{I - jQ} y^3.$$

Since we are still analyzing in the case that $|x| = 1$, we know that $|y| = \left|\frac{1}{x}\right| = 1$. So,

$$|F(x)| = \left| \frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} \right|$$

$$|G(x)| \leq 1 + \left| \frac{I + jQ}{I - jQ} \right| \left| \frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} \right| + \left| \frac{I + jQ}{I - jQ} \right|.$$

Continuing with the assumptions that Q leads I and distortion is not severe, we find that

$$|F(x)| - |G(x)| \geq \left| \frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} \right| \left(1 - \left| \frac{I + jQ}{I - jQ} \right| \right) - \left| \frac{I + jQ}{I - jQ} \right| - 1$$

$$> 0.$$

Because $F(y)$ has two identical roots $y = 0$ inside the unit circle $|y| < 1$, Rouché's Theorem tells us that $Q(y) = F(y) + G(y)$ should have two zero roots for $|y| < 1$.

Since $Q(y) = y^3 L\left(\frac{1}{y}\right) = \frac{1}{x^3} L(x)$, we know that $L(x)$ will have exactly two zero roots for $|x| > 1$.

Thus, we have located the three roots of $L(x)$. We know that $|x_1| < 1, |x_2| > 1, |x_3| > 1$.

§4.3 Approximating the roots

To approximate the roots, we take advantage of the fact that $|x_1| < 1$. We can rewrite $L(x) = 0$:

$$x^3 + \frac{I + jQ}{I - jQ} \cdot \frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} x^2 + \frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} x + \frac{I + jQ}{I - jQ} = 0$$

$$\frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} x \left(\frac{I + jQ}{I - jQ} x + 1 \right) + x^3 + \frac{I + jQ}{I - jQ} = 0.$$

Dividing both sides by $x \left(\frac{I + jQ}{I - jQ} x + 1 \right)$, we get

$$\frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2} + \frac{x^3 + \frac{I + jQ}{I - jQ}}{x \left(\frac{I + jQ}{I - jQ} x + 1 \right)} = 0$$

$$\frac{x^3}{x \left(\frac{I + jQ}{I - jQ} x + 1 \right)} + \frac{\frac{I + jQ}{I - jQ}}{x \left(\frac{I + jQ}{I - jQ} x + 1 \right)} = -\frac{4 + 3\delta_2 A_0^2}{\delta_2 A_0^2}.$$

After further simplification, we get

$$\frac{x^2}{\left(\frac{I+jQ}{I-jQ}x+1\right)} + \frac{x^{-1}\frac{I+jQ}{I-jQ}}{\left(\frac{I+jQ}{I-jQ}x+1\right)} = -\frac{4+3\delta_2A_0^2}{\delta_2A_0^2}$$

$$\frac{x^2}{\left(\frac{I+jQ}{I-jQ}x+1\right)} + \frac{x^{-1}}{\left(x+\frac{I-jQ}{I+jQ}\right)} = -\frac{4+3\delta_2A_0^2}{\delta_2A_0^2}.$$

Now, considering the root $|x_1| < 1$, we know that

$$\left| \frac{x_1^2}{\left(\frac{I+jQ}{I-jQ}x_1+1\right)} \right| \ll \left| \frac{x_1^{-1}}{\left(x_1+\frac{I-jQ}{I+jQ}\right)} \right|.$$

Thus, the second term $\frac{x_1^{-1}}{\left(x_1+\frac{I-jQ}{I+jQ}\right)}$ will dominate and we can make the approximation

$$-\frac{4+3\delta_2A_0^2}{\delta_2A_0^2} \approx \frac{x_1^{-1}}{\left(x_1+\frac{I-jQ}{I+jQ}\right)} \approx \frac{x_1^{-1}}{\left(\frac{I-jQ}{I+jQ}\right)}.$$

So we can approximate x_1 pretty well by $x_1 \approx -\frac{\delta_2A_0^2}{4+3\delta_2A_0^2} \cdot \frac{I+jQ}{I-jQ}$.

By Vieta's, we know that $x_1x_2x_3 = -\frac{I+jQ}{I-jQ}$. Thus, we can approximate the product x_2x_3 by

$$x_2x_3 \approx \frac{4+3\delta_2A_0^2}{\delta_2A_0^2}.$$

§4.4 Continuing the Analysis on Phase Shifter Gain Under Nonidealities

Recall that we found G_{norm} to be

$$G_{\text{norm}} = e^{j3\theta} \left(\frac{I-jQ}{2} \frac{\delta_2A_0^2}{4} \right)^N [e^{-j2\theta} + (-x_1)^N] [e^{-j2\theta} + (-x_2)^N] [e^{-j2\theta} + (-x_3)^N].$$

For large N , we can make the following approximations reasonably:

$$e^{-j2\theta} + (-x_1)^N \approx e^{-j2\theta}$$

$$[e^{-j2\theta} + (-x_2)^N] [e^{-j2\theta} + (-x_3)^N] \approx (x_2x_3)^N \approx \left(\frac{4+3\delta_2A_0^2}{\delta_2A_0^2} \right)^N.$$

Thus, for sufficiently large N we can rewrite G_{norm} as

$$G_{\text{norm}} \approx e^{j3\theta} \left(\frac{I-jQ}{2} \frac{\delta_2A_0^2}{4} \right)^N e^{-j2\theta} \left(\frac{4+3\delta_2A_0^2}{\delta_2A_0^2} \right)^N$$

$$\approx e^{j\theta} \left(\frac{I-jQ}{2} \frac{4+3\delta_2A_0^2}{4} \right)^N.$$

Thus, we can see that as N increases, the magnitude of G_{norm} doesn't rely on θ and the output phase is linearly related to θ .

The response will, therefore, converge to that of an ideal phase shifter.

§4.5 Higher order Taylor Expansions

Recall that we found the second order Taylor expansion on δ to achieve these results.

If we used an M th order Taylor expansion instead, we can proceed in the same fashion as when we approximated $\delta(A, f) \approx \delta_2 A^2$. The results will not change as we can simply change $\delta_2 A^2$ in all our expressions to whatever the Taylor expansion becomes.

§5 Over-Coupling for Bandwidth Extension

Traditionally, Lange couplers are set to equal coupling with $C = \sqrt{2}/2$ to get equal-amplitude quadrature signals. However, the proposed SCVM architecture allows the coupling coefficient to be higher.

Plugging in the values of I and Q for the phase error found in [subsection 3.4](#), we find

$$\begin{aligned}
 |\Delta_\theta|_{\max} &= \arcsin \left(\left| \frac{I + jQ}{I - jQ} \right|^N \right) \\
 &= \arcsin \left(\left| \frac{\sqrt{1-C^2} - C \sin\left(\frac{\pi f}{2f_c}\right)}{\sqrt{1-C^2} + C \sin\left(\frac{\pi f}{2f_c}\right)} \right|^N \right) \\
 &= \arcsin \left(\left| \frac{1 - \frac{C \sin\left(\frac{\pi f}{2f_c}\right)}{\sqrt{1-C^2}}}{1 + \frac{C \sin\left(\frac{\pi f}{2f_c}\right)}{\sqrt{1-C^2}}} \right|^N \right) \\
 &= \arcsin \left(\left| \frac{2}{1 + \frac{C}{\sqrt{1-C^2}} \sin\left(\frac{\pi f}{2f_c}\right)} - 1 \right|^N \right).
 \end{aligned}$$

§5.1 Equal Coupling

Define $K = \frac{C}{\sqrt{1-C^2}}$ for the sake of simplicity. From the values for I and Q presented in [section 2](#), we can find that $K = |Q/I|$. Traditionally, $K = 1$, and the minimum operation frequency for a maximum acceptable phase error $\Delta\theta_{\text{goal}}$ would be

$$\begin{aligned}
 |\Delta_\theta|_{\text{goal}} &= \arcsin \left(\left| \frac{2}{1 + K \sin\left(\frac{\pi f}{2f_c}\right)} - 1 \right|^N \right) \\
 \sqrt[N]{\sin(|\Delta_\theta|_{\text{goal}})} &= \frac{2}{1 + \sin\left(\frac{\pi f}{2f_c}\right)} - 1 \\
 1 + \sin\left(\frac{\pi f}{2f_c}\right) &= \frac{2}{1 + \sqrt[N]{\sin(|\Delta_\theta|_{\text{goal}})}} \\
 f &= \frac{2f_c}{\pi} \arcsin \left(\frac{2}{1 + \sqrt[N]{\sin(|\Delta_\theta|_{\text{goal}})}} - 1 \right).
 \end{aligned}$$

However, K no longer needs to be 1 in the SCVM architecture because of the exponential error reduction.

§5.2 Overcoupling

Below is a figure I generated with Python on the impact of K on phase error for different numbers of stages.

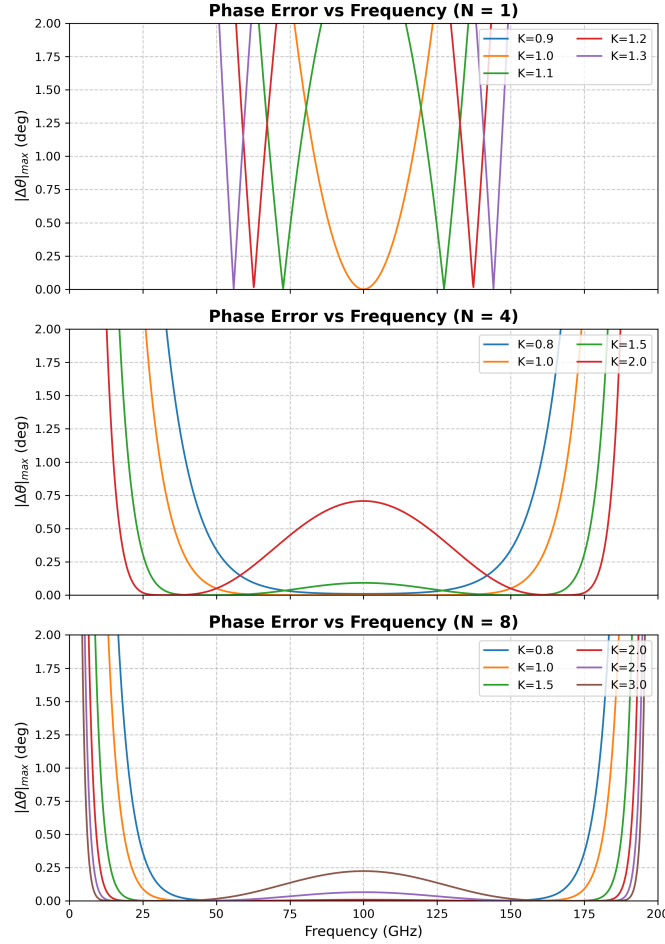


Figure 2: Impact of K on PS for given stage numbers when $f_c = 100$ GHz

Note that in the $N = 1$ case, designing $K = 1$, lower phase errors can be achieved in a small bandwidth around f_c . However, in the $N = 4$ case, the figure shows that a large bandwidth can be achieved even for $K \neq 1$. In general, we can find the minimum operating frequency for arbitrary K and a maximum acceptable phase errors as follows:

$$|\Delta_\theta|_{\text{goal}} = \arcsin \left(\left| \frac{2}{1 + K \sin \left(\frac{\pi f}{2f_c} \right)} - 1 \right|^N \right)$$

$$\sqrt[N]{(\sin(|\Delta_\theta|_{\text{goal}}))} = \frac{2}{1 + K \sin \left(\frac{\pi f}{2f_c} \right)} - 1$$

Adding one on both sides and multiplying, we find

$$1 + K \sin\left(\frac{\pi f}{2f_c}\right) = \frac{2}{1 + \sqrt[N]{\sin(|\Delta\theta|_{\text{goal}})}}$$

$$f = \frac{2f_c}{\pi} \arcsin\left(\frac{1}{K} \left(\frac{2}{1 + \sqrt[N]{\sin(|\Delta\theta|_{\text{goal}})}} - 1\right)\right).$$

This equation demonstrates that as K increases, the minimum value for f will decrease. Furthermore, as N increases, f will also decrease.

§6 Summary

Below, I have curated a list of the main ideas in the mathematical modeling for this paper.

1. By setting the each of the phase shifters a root of unity apart, the symmetry allows the authors to create "self-compensation".
2. The authors turn $I \cos(\alpha_k) + Q \sin(\alpha_k)$ into a sum of two exponentials using Euler's formula to work with complex numbers, which are helpful in this case.
3. A good idea is presented in [subsection 3.2](#), where the authors extract a product of terms that consist of a constant added with some magnitude multiplied by $e^{-2j\alpha_k}$. Then, Vieta's is applied and the product can be cleared to the sum of two terms. This idea is utilized twice.
4. Symmetry can again be seen in [section 4](#), where the symmetry of the VGA is exploited to create an even function δ with respect to A . This simplifies the Taylor Expansion.
5. To get rid of an ugly product, the authors defined $L(x)$ to simplify the work greatly.

§7 Paper's Errata

Below is a list of small errata in the paper.

1. Equation (6) is not quite right for odd N . However, it is, in essence, correct. Use my method for simplicity.
2. Figure 5 in the paper incorrectly plots gain error instead of phase error for $N = 1$.
3. Equation (58) is missing an x . It should be $\frac{4+3\delta_2 A_0^2}{\delta_2 A_0^2} x \left(\frac{I+jQ}{I-jQ} x + 1\right)$ in that first term.